EchoYeast hardware module — final purchase list

Updated 2026-10-19 · Parts list for the current revision

No.	Category	Item	Specification / model	Purpose
1	Structural parts & printing consumables	3D printing filament (PLA)	PLA Basic, white 1000 g + black 1000 g (without spool)	FDM printing of the collection cup, the heater-block body and the three plug-in modules
2	Temperature control	Heat-resistant thermal silicone pad	200 × 400 × 0.3 mm, 6 W/(m·K)	Thermal interface between the heating film and the heater block; cut to size
3	Temperature control	PI heating film	60 × 70 mm, 12 V / 25 W	Bonded to the underside of the heater block as the heating element
4	Temperature control	MAX31865 module + PT100	MAX31865 converter module + PT100 platinum RTD (3-wire)	Temperature acquisition: PT100 resistance converted to a digital temperature by a 16-bit ADC
5	Temperature control	KSD9700 thermal cut-off	60 °C, 250 V / 10 A, metal case, normally closed	Hardware-level over-temperature protection; in series with the heater so it physically cuts the circuit
6	Temperature control	MOSFET driver module	IRF520	Takes the PWM signal and drives the heating film to implement PID power control
7	Temperature control	Cooling fan	40 × 40 × 10 mm, 12 V, quiet	Mounted on the heat sink to keep the electronics underneath and the block from overheating
8	Temperature control	Heat sink	Aluminium, 70 × 36 × 100 mm	Dissipates heat downwards and protects the circuit underneath
9	Optics	LED light source	485 nm high-brightness LED	Excitation light source; spares kept for wavelength margin
10	Optics	LED constant-current driver	Adjustable constant-current driver board, 350 / 500 mA	Ordered, but measured at a fixed 14.7 mA with no effect from the trimmer; withdrawn and replaced by a 200 Ω series resistor (item 25)
11	Optics	Photodiode	BPW34, through-hole package	Detects the 528 nm fluorescence at the receiver end; spares kept
12	Optics	Narrow-band filters	Ø10 mm, 30 nm bandwidth, OD3; 485 nm × 2 + 528 nm × 2	One at each end separates the excitation light from the fluorescence spectrum
13	Optics	Op-amp	OPA350UA (SOIC-8)	Forms the transimpedance amplifier with the photodiode and outputs a measurable voltage
14	Optics	Light shield (black heat-shrink tubing)	12 mm bore, 10× thick wall	Wraps the emitter and receiver plugs to exclude ambient light
15	Optics	Focusing lenses for emitter and receiver	Matched to the 485 nm emitter and the 528 nm receiver (newly ordered)	Focus the LED output onto the centre of the vessel and the fluorescence onto the photodiode active area
16	Electronics & signal	Microcontroller	Arduino Nano V4 (board + USB cable)	Timing control, ADC readout and the PID temperature algorithm
17	Electronics & signal	ADS1115 high-precision ADC	ADS1115 module, 16-bit, 4-channel differential, I²C (newly ordered)	Reads the amplified photodiode voltage with high precision
18	Electronics & signal	SOT89 adapter board	SOT89 to DIP through-hole adapter (newly ordered)	Lets SOT89 SMD parts be soldered through-hole
19	Electronics & signal	SOT223 adapter board	SOT223 to DIP through-hole adapter (newly ordered)	Lets SOT223 SMD parts be soldered through-hole
20	Power & wiring	Power adapter	24 V / 2 A (48 W), with mains lead	Supplies the instrument; 24 V bus

No.	Category	Item	Specification / model	Purpose
21	Power & wiring	LM2596 buck module	LM2596S DC-DC adjustable step-down	Steps 24 V down to 12 V (heating film, fan) and 5 V (logic)
22	Power & wiring	Breadboard / protoboard	Single-sided, rosin-coated green, 12 × 18 cm	Prototyping during debugging, then soldered permanently
23	Power & wiring	Jumper wires	21 cm, male–male / male–female / female–female, 120 pcs total	Temporary connections between boards and modules
24	Power & wiring	Electrolytic capacitors	16 V / 100 µF, 5 × 11 mm	Supply filtering; suppresses switching noise
25	Power & wiring	Resistor kit	1/4 W metal film, 25 common values, 500 pcs total	Current limiting, voltage division and transimpedance feedback
26	Power & wiring	Indicator LEDs	5 mm red / green / yellow, 60 pcs total	Power and run-status indication
27	Power & wiring	Terminal blocks	KF301-2P × 10 + 3P × 10, 5.08 mm	Power input and load wiring
28	Power & wiring	30 AWG silicone wire	Extra-flexible silicone wire for wiring inside the plugs and for signals (newly ordered)	Fine wiring inside the plugs keeps them compact and flexible
29	Tools & consumables	Thermal silicone adhesive	Kafuter K-5204K, fast-curing, white	Fixes the heating film, PT100 and heat sink; also seals against light
30	Tools & consumables	Heat-resistant double-sided tape	PI polyimide tape, 0.1 mm thick, 3 mm × 10 m	Holds the heating film onto the heater-block surface
31	Tools & consumables	Silicone wire	22 AWG red + black, rated ≥ 105 °C	Power wiring for the heater and supply circuits
32	Tools & consumables	Desoldering wick	For desoldering and rework (newly ordered)	Rework and clearing of mis-soldered joints
33	Tools & consumables	Multimeter	UNI-T UT33A, auto-ranging	Voltage, current, continuity and resistance measurement
34	Tools & consumables	Soldering station (936)	60 W adjustable constant temperature, standard model	Soldering the circuits and plug-in modules
35	Tools & consumables	Hot-melt glue gun	40 W, with glue sticks	Securing components and sealing light leaks
36	Tools & consumables	Solder wire	Lead-free, 0.8 mm, rosin core	Soldering consumable
37	Optics	Transimpedance feedback resistor	680 kΩ, 1/4 W metal film, 1%	Sets the transimpedance gain (U = I × R): 1 µA becomes 0.68 V
38	Optics	Ceramic capacitor	0.1 µF, 50 V, X7R	C10 damps the transimpedance stage (≈2.3 Hz); C9 decouples the op-amp at pins 7 and 4
39	Optics	Pin header and socket pair	2.54 mm, 1×2P, header + socket	Makes the LED and the BPW34 assembly plug-in; a reversed polarity is cured by rotating the header

Summary by category

Category	Items
Structural parts & printing consumables	1
Temperature control	7
Optics	10
Electronics & signal	4
Power & wiring	9
Tools & consumables	8
Total	39

Purchase links (items already ordered)

Item	Link
3D printing filament (PLA)	https://e.tb.cn/h.8D2q9Wgd9FICQIA?tk=4wliTOpQjq4
Heat-resistant thermal silicone pad	https://e.tb.cn/h.8D2u35e03HzrnKQ?tk=ZUlpTOpnJNj
PI heating film	https://e.tb.cn/h.8DdYQSZ9I7Gn80H?tk=q5L3TOpvEUn
MAX31865 module + PT100	https://e.tb.cn/h.8Eo6m9Y1yBcqsCB?tk=0hdiTOpzeHk
KSD9700 thermal cut-off	https://e.tb.cn/h.8EoisstnhcDy4WH?tk=nJM9TOKZxsu
MOSFET driver module	https://e.tb.cn/h.8FTxa7A424Jc4TE?tk=Gf2cTOKc66T
Cooling fan	https://e.tb.cn/h.8FTCMdiqBypUWOx?tk=15evTOKX21L
Heat sink	https://e.tb.cn/h.8vh4852tGrpl6rI?tk=MU7NTOKf3Em
LED light source	https://e.tb.cn/h.8vhUy8oFTunUWog?tk=3NKaTOKg7rV
LED constant-current driver	https://e.tb.cn/h.8EoqXI9Ckjpxpg9?tk=B6LBTOK6QOH
Photodiode	https://e.tb.cn/h.8vhSus9evlimDIM?tk=drwcTOK7u8T
Narrow-band filters	https://e.tb.cn/h.8F6ngurIWsopc7N?tk=susnTOKFY6Y
Op-amp	https://e.tb.cn/h.8DWYYiqtEKvqeLx?tk=O7y0TOKxbDm
Light shield (black heat-shrink tubing)	https://e.tb.cn/h.8ELgpJKO9UjsSkU?tk=7ORATOKzjE0
Microcontroller	https://e.tb.cn/h.8ELBloVll81AYYP?tk=ZfVFTOqJpe2
Power adapter	https://e.tb.cn/h.8F6tQHNABFojNUa?tk=Vds9TOKARZj
LM2596 buck module	https://e.tb.cn/h.8DWdMqYnWoZeTiV?tk=qyQATOqZxlt
Breadboard / protoboard	https://e.tb.cn/h.8DWeeX1yDMySMdW?tk=gWEGTOqbyTD
Jumper wires	https://e.tb.cn/h.8vS0UJczgCS6B9I?tk=UVU1TOqX7Bc

Item	Link
Electrolytic capacitors	https://e.tb.cn/h.8DWTRQb5qqQUXFB?tk=XnXFTOq3l20
Resistor kit	https://e.tb.cn/h.8Fh0TDbu7fD8gpo?tk=QrEITOq4cHV
Indicator LEDs	https://e.tb.cn/h.8ELnMVRucfBtMhc?tk=ielnTOqUEXm
Terminal blocks	https://e.tb.cn/h.8ELo5JbhvB3KreD?tk=uO7KTOqgam2
Thermal silicone adhesive	https://e.tb.cn/h.8Fhd5G7Prf3J7S0?tk=t5SDTOqT7Sk
Heat-resistant double-sided tape	https://e.tb.cn/h.8DWQ2ns5GLIGnhN?tk=KnPSTOqSk57
Silicone wire	https://e.tb.cn/h.8ELJRdCyEZp1FDH?tk=5qyfTOq7oBz
Multimeter	https://e.tb.cn/h.8Fh5C8Nsg2KiLLM?tk=q5DbTOqjBmh
Soldering station (936)	https://e.tb.cn/h.8DWpiXAZNefdTRU?tk=A0vmTOqObkV
Hot-melt glue gun	https://e.tb.cn/h.8ELFziG8HafPdXK?tk=Mr0HTOqNcMG
Solder wire	https://e.tb.cn/h.8FhLCVc3yUmjPZa?tk=Lg8xTOquJTR